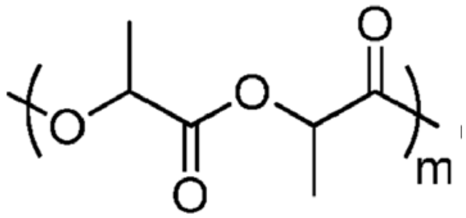
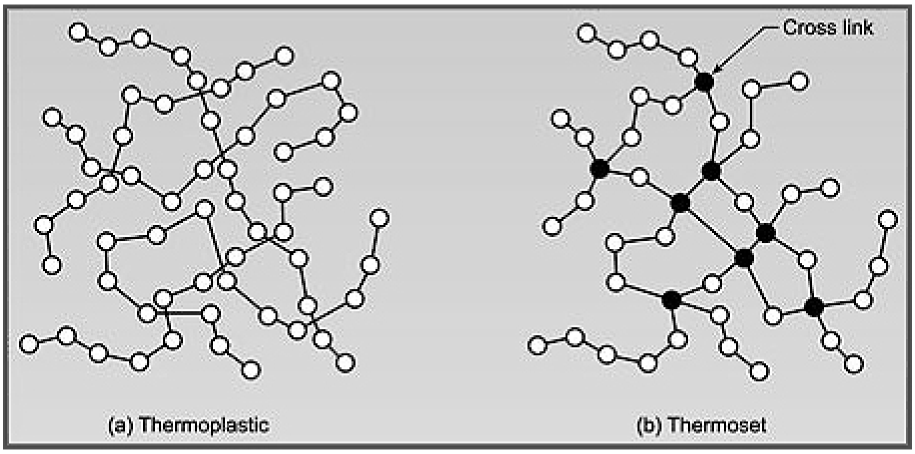
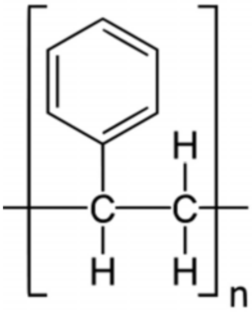
1	(a)	**W**: $SiCl_4$ $SiCl_4 + 2H2O \rightarrow SiO_2 + 4HCl$ **X**: P_4O_{10} and P_4O_6 $P_4O_{10}(s) + 6H_2O(l) \rightarrow 4H_3PO_4(aq)$ $P_4O_6(s) + 6H_2O(l) \rightarrow 4H_3PO_3(aq)$	**[3]**
	(b) (i)	$Al_2O_3 + 2NaOH \rightarrow 2Na[Al(OH)_4]^- + H_2O$ OR $Al_2O_3 + 2OH^- \rightarrow 2[Al(OH)_4]^- + H_2O$	**[1]**
	(b) (ii)	On addition of $HCl(aq)$, a white ppt is formed. Ppt would dissolve in excess $HCl(aq)$ to give a colourless solution.	**[2]**
	(c)	Group 17 elements have simple covalent structure, between discrete molecules there exist weak instantaneous dipole induced dipole interactions. Down group 17, the molecular size of halogen increases, increasing the size of electron cloud and the instantaneous dipole induced dipole interactions. Thus, group 17 elements are less volatile down the group.	**[2]**
	(d)	The 3p electron to be removed from **W** is at a higher energy level compared to the 3s electron to be removed from **Z**, hence the p electron is less strongly attracted to the nucleus and require less energy to remove.	**[1]**
			[Total: 9]

2	(a)	Nanostructure is a structure with one or more dimensions in the size range 1nm to 100nm.		**[1]**
	(b)	On coming in contact with any surface, each spatula deforms to enabling maximum contact with the wall(substrate) surface**,** forming instantaneous dipole induced dipole interactions.As spatulae are nanostructure which possessing high surface to volume ratio and there are billions of spatulae per gecko**,** thus creating a huge collective surface area of contact. The culmulative spatula-wall instantaneous dipole induced dipole interactions generated is therefore translated to enormous attractive/adhesive forces capable of supporting many times the animal's body weight.		**[3]**
	(c)	(i)	The nanostructure on lotus leaf increases the total surface area of the leaf but reduces the contact area for the debris with the leaf. With a reduced contact area, the hydrophobic interaction between debris and leaf is largely reduced, making it easy for water droplets to remove debris off the leaf giving rise to self cleaning properties of lotus leaf.	**[2]**

		(ii)	The <u>increased surface area to volume ratio on lotus leaf will enhance the hydrophobic interactions</u> between the leaf and organic solvent. The <u>Organic solvent will wet the leaf/ not form droplets on leaf/ closely bind to leaf surface</u> and thus the self-cleaning property will not be exhibited with organic solvent.	[2]
				[Total: 8]

3	**(a)**	Regent and conditions: acidified $KMnO_4$ / $K_2Cr_2O_7$, heat under reflux $CH_3COCOOH$		**[2]**
	(b)	$CH_3CH(OH)CH_3OH$		**[1]**
	(c)	 Reagent and Condition: a few drops of conc sulfuric acid, heat under reflux Type of polymerisation: condensation polymerisation		**[3]**
	(d)	**(i)**	 Thermoplastic has little or no cross linking where thermosetting plastics has cross linking	**[3]**
		(ii)	Softening behaviour/lower degree of rigidity/ lower strength in material	**[1]**

		(iii)	 Type of polymerisation: free radical polymerisation	[2]
	(d)	The weaker interactions between polymer chains in thermoplastics break down at much lower temperatures than the chemical bonds between monomers, allowing thermoplastics to be recycled indefinitely until the polymers are broken down to the point that the material loses structural integrity. Thermosets, on the other hand, can only be heated once which is during the thermosetting transformation to form the hard plastic. The transformation is an irreversible change. If a thermoset plastic is heated for a second time to a high temperature, the polymer chain will degrade and the plastic will burn instead of melt. This results in a lower recyclability of thermosets.		[2]
				[Total: 14]

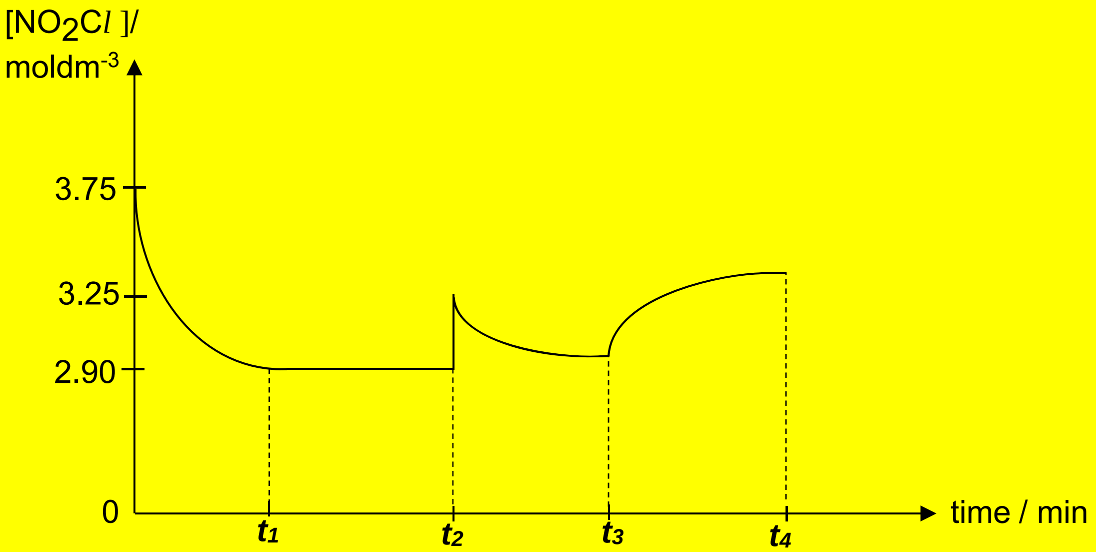
4 (a)

Initial conc of NO = $\frac{1}{4} \times 5 = 1.25$ atm

Initial conc of NO_2Cl = 5 – 1.25 = 3.75 atm

	NO_2Cl	+	NO	⇌	NOC*l*	+	NO_2
Initial / $moldm^{-3}$	3.75		1.25		0		0
Δ / $moldm^{-3}$	–0.85		–0.85		+0.85		+0.85
Eqm / $moldm^{-3}$	2.90		0.40		0.85		0.85

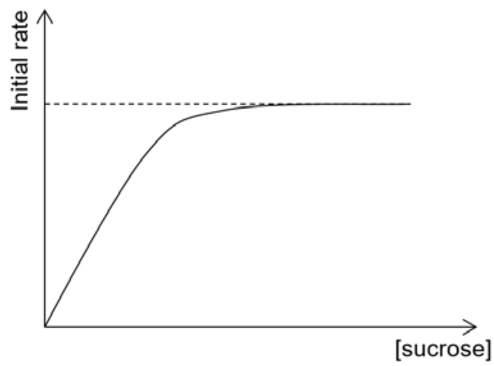
$K_c = \frac{(0.85)(0.85)}{(2.90)(0.40)} = \underline{0.623}$ (no units) **[3]**

	(b)			[2]
	(c)			[1]
	(d)		Bent shape; accept 100° to <120°	[2]
				[Total: 8]

5	(a)		An acid (H_2SO_4) is a proton donor. A base (HNO_3) is a proton acceptor. The conjugate acid of a base ($H_2NO_3^+$) is the species formed from the base accepting a proton. The conjugate base of an acid (HSO_4^-) is the species formed from the acid donating a proton.	[2]
	(b)	(i)	No. of moles of H_2SO_4 present = $\frac{25.0}{1000} \times 0.0210 = 0.000525$ mol No. of moles of NH_3 required = $2 \times 0.000525 = 0.00105$ mol Volume of NH_3 required = $\frac{0.00105}{0.0500} \times 1000 = \underline{21.00\ cm^3}$	[1]
		(ii)	No. of moles of NH_3 added = $\frac{10.00}{1000} \times 0.0500 = 0.000500$ mol No. of moles of H_2SO_4 reacted = $\frac{0.000500}{2} = \underline{0.000250\ mol}$ No. of moles of H_2SO_4 initially = $\frac{25.0}{1000} \times 0.0210 = 0.000525$ mol No. of moles of H_2SO_4 remaining = 0.000525 – 0.000250 = $\underline{0.000275\ mol}$	[2]

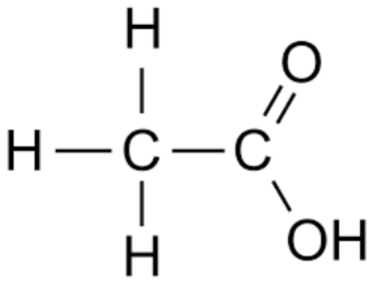
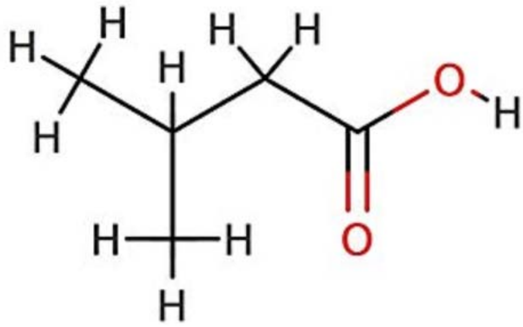
		(iii)	No. of moles of H^+ remaining = 2 × 0.000275 = 0.000550 mol $[H^+] = \frac{0.000550}{(\frac{35.0}{1000})}$ = $\underline{0.0157 \text{ mol dm}^{-3}}$ pH = –lg(0.0157) = $\underline{1.80}$	**[2]**
	(c)	**(i)**	$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}$	**[1]**
		(ii)	pOH = 14 – 9.5 = 4.5 4.5 = –lg(1.8 × 10^{-5}) + lg($\frac{[salt]}{0.0500}$) $[NH_4Cl]$ = $\underline{0.0285 \text{ mol dm}^{-3}}$ Mass of NH_4Cl = 0.0285 × 1.00 × (14.0 + 4 × 1.0 + 35.5) = $\underline{1.52 \text{ g}}$	**[2]**
	(d)	**(i)**	H_2CO_3 and HCO_3^-	**[1]**
		(ii)	$HCO_3^- + H^+ \rightarrow H_2CO_3$ Lactic acid causes the $[H^+]$ in blood to increase. The HCO_3^- present <u>neutralises all the additional H^+</u>, allowing the pH to remain relatively constant.	**[1]**
				[Total: 12]

6	**(a)**	**(i)**	Comparing Experiments 1 and 2, When the volume/concentration of HC*l* increases to 1.5 times, rate increases to 1.5 times. Hence, reaction is first order with respect to HC*l*. Comparing Experiments 1 and 3, When the volume/concentration of sucrose doubles, rate doubles. Hence, reaction is first order with respect to sucrose.	**[2]**
		(ii)	Rate = k [sucrose] [HC*l*] From Experiment 1, [sucrose] = 0.850 × $\frac{20}{50}$ = 0.340 mol dm^{-3} [HC*l*] = 1.23 × $\frac{30}{50}$ = 0.738 mol dm^{-3} $k = \frac{1.77 \times 10^{-3}}{(0.340)(0.738)}$ = 7.05 × 10^{-3} mol^{-1} dm^3 min^{-1}	**[2]**

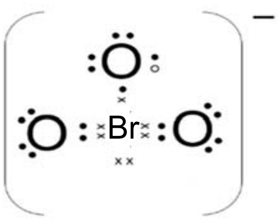
		(iii)	[sucrose] = 0.850 × $\frac{40}{70}$ = 0.486 mol dm^{-3} [HC*l*] = 1.23 × $\frac{20}{70}$ = 0.351 mol dm^{-3} Both concentrations for **[1]** Initial rate of reaction = 7.05 × 10^{-3} × 0.486 × 0.351 = 1.20 × 10^{-3} mol dm^{-3} min^{-1}	**[2]**
	(b)	**(i)**		**[1]**
		(ii)	When [sucrose] is low, reaction is first order with respect to sucrose due to the availability of active sites on the enzyme molecules for binding. When [sucrose] is high, all active sites on the enzyme molecules are occupied. The rate of reaction then is independent of [sucrose] and the reaction is zero order with respect to sucrose.	**[2]**
			[Total: 9]	

Section B

7	(a)	$1s^2 2s^2 2p^6 3s^2 3p^1$ Fourth electron is removed from an inner principal quantum shell which is closer to the nucleus while the first three electrons are removed from the outermost shell. The fourth electron in the inner shell (2p subshell) also experiences a greater effective nuclear charge compared to the electrons in the outermost shell (3s and 3p subshells). Hence, the fourth electron is more strongly attracted to the nucleus and requires more energy to be removed.	[3]
	(b)	Al has electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^1$. Si has electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^2$. The 1st ionisation energy for both Al and Si involve the removal of a 3p electron. Si has a greater nuclear charge compared to Al, while the shielding effect experienced by the 3p electron of Si is approximately the same as that experienced by the 3p electron of Al as they are shielded by the same number of inner shells of electrons. The first 3p electron in Si experiences a greater effective nuclear charge than Al. The atomic radius of Si is also smaller than that of Al. Hence, more energy is required to remove electrons from Si atoms compared to Al atoms.	[4]
	(c)	B, Al and Ga are Group 13 elements. Down the group, atomic radius increases. Screening effect increases due to increase in number of electronic shells, which largely cancels out the increase in nuclear charge. Hence the valence electrons become increasingly less attracted by the positive nucleus and less energy is required to remove them.	[2]
	(d)	Test: Bromine in CCl_4 Observation: orange-red solution decolourise to form colourless solution	[2]
	(e)	Molecular formula of geraniol = $C_{10}H_{18}O$ % C by mass = [12.0x10 /(12.0x10 + 1.0x18 + 16)] x100% = 77.9%	[2]
	(f)	(i) H_2SO_4(aq) / NaOH(aq), heat under reflux	[1]

		(ii)	  [accept salt if condition is NaOH]	[2]
		(iii)	1710-1750 cm^{-1}	[1]
		(iv)	3580-3650 cm^{-1}	[1]
		(v)	q/m ratio for geranyl acetate = +1 / 196 q/m ratio for geranyl isovelerate = +1 / 238 angle of deflection = (+1/238) (20) / (+1/196) = 16.5°	[2]
				[Total: 20]

8	(a)	(i)	An indicator is not required as the green Fe^{2+} will be oxidised to form yellow Fe^{3+}.	[1]
		(ii)	No. of moles of Fe^{2+} = $\frac{3.420}{55.8+32.1+16.0\times4}$ = <u>0.0225 mol</u> [Fe^{2+}] in **FA 1** = $\frac{0.0225}{250/1000}$ = <u>0.0900 mol dm^{-3}</u>	[2]
		(iii)	No. of moles of Fe^{2+} in 25.0 cm^3 = $\frac{25.0}{1000} \times 0.0900$ = <u>0.00225 mol</u>	[1]
		(iv)	Average volume of $NaBrO_3$ = $\frac{(25.60-0.00)+(38.20-12.50)}{2}$ = <u>25.65 cm^3</u> No. of moles of BrO_3^- required = $\frac{25.65}{1000} \times 0.044$ = <u>0.00113 mol</u>	[2]
		(v)	$Fe^{2+} \equiv e^-$ No. of moles of e^- lost by Fe^{2+} = 0.00225 mol No. of moles of e^- gained by 1 mole of BrO_3^- = $\frac{0.00225}{0.00113}$ = 1.99 mol = <u>2 mol</u> Final oxidation state of bromine = +5 – 2 = <u>+3</u>	[2]
		(vi)	BrO_2^- $2Fe^{2+} + 2H^+ + BrO_3^- \rightarrow 2Fe^{3+} + BrO_2^- + H_2O$	[2]

	(b)	(i)		[1]
		(ii)	Trigonal pyramidal 107°	[2]
		(iii)	The lone pair-bond pair repulsion in BrO_3^- is weaker than the lone pair-lone pair repulsion in water.	[1]
		(iv)	Br_2 has a simple covalent structure while NaBr has a giant ionic lattice structure. Less energy is required to overcome the weaker instantaneous dipole-induced dipole (id-id) interactions between Br_2 molecules than the stronger electrostatic attraction between Na^+ and Br^-. Both Br_2 and BrO_2 have simple covalent structure. Less energy is required to overcome the weaker id-id interactions between Br_2 molecules than the stronger permanent dipole-permanent dipole interactions between BrO_2 molecules. Both $NaBrO_3$ and NaBr have giant ionic lattice structure. BrO_3^- is larger than Br^-. Less energy is required to overcome the weaker electrostatic attraction between Na^+ and BrO_3^- than that between Na^+ and Br^-.	[4]
	(c)	At low temperature, intermolecular forces of attraction can form easily between gaseous particles for the gas to liquefy. At high pressure, gaseous particles are closer together for the gas to liquefy.		[2]
		[Total: 20]		